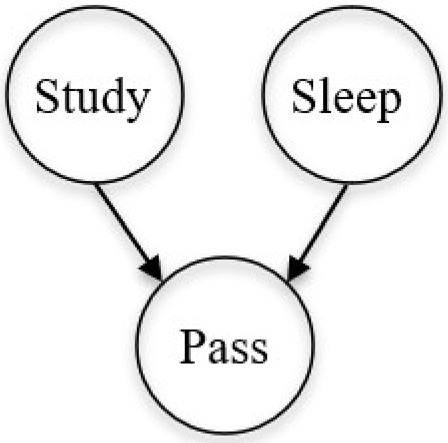
Bayesian Networks

Lab 8

Exercise 1: Learning From Data (Alarm.py)

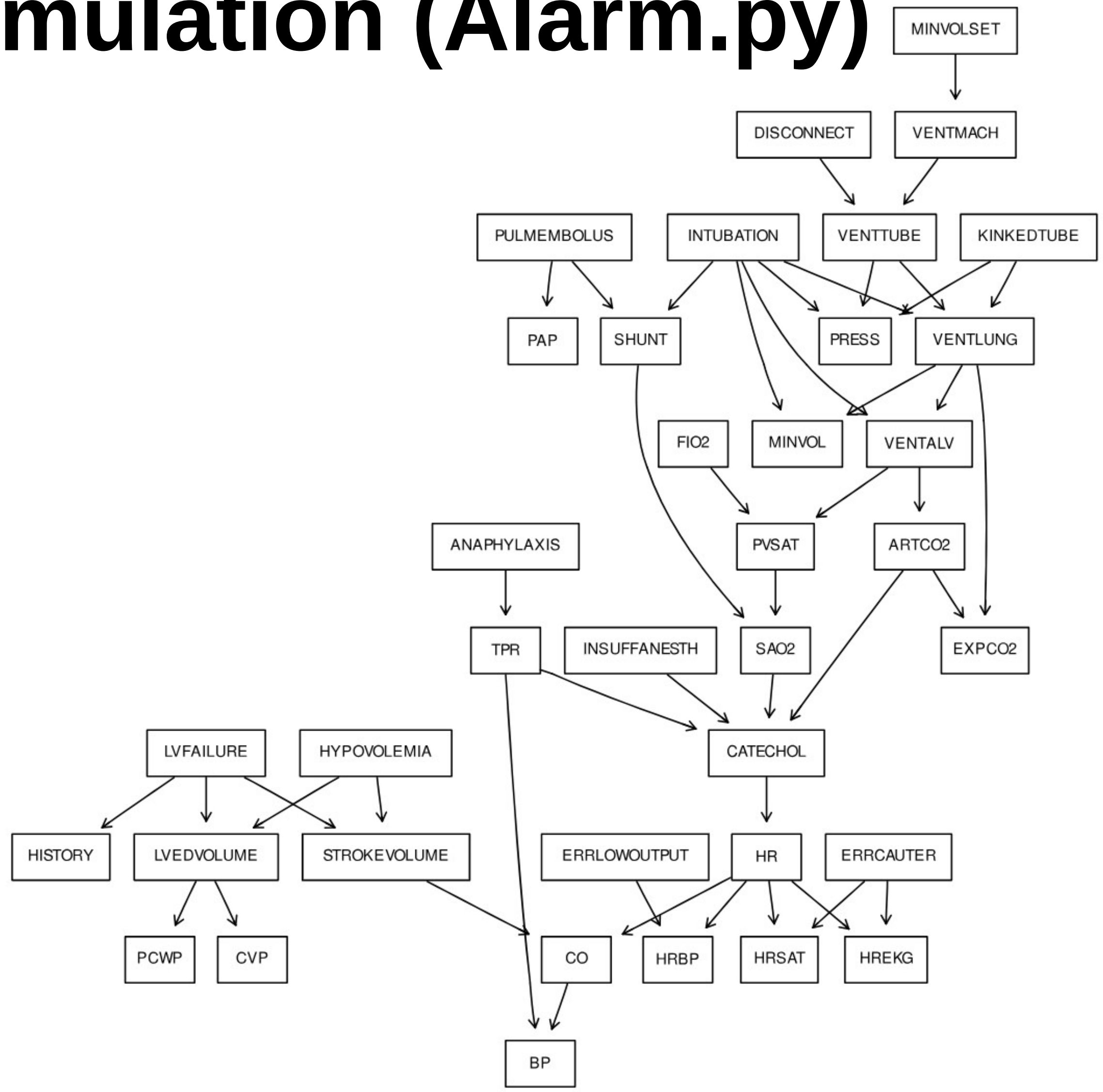
→ Given a dataset consisting of three binary variables (Sleep, Study, Pass), and given a DAG, learn the BN's parameters from the following data:

Study	Sleep	Pass
1	1	1
1	1	1
1	0	1
1	0	0
0	1	0
0	1	0
0	0	0
0	0	0
1	1	1
0	1	0



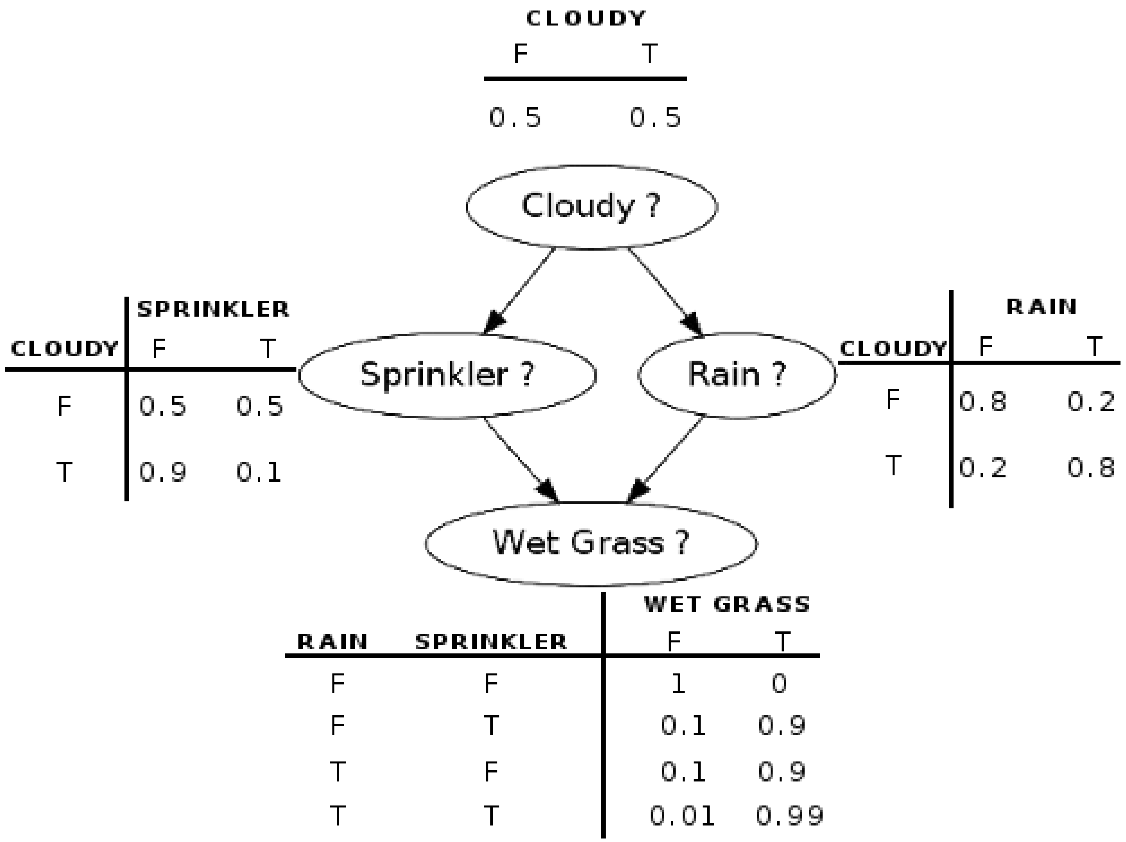
Exercise 2: Learning From Simulation (Alarm.py)

- Load the existing Alarm network from bnlearn. The ALARM ("A Logical Alarm Reduction Mechanism") is a Bayesian network designed to provide an alarm message system for patient monitoring. Simulate 100 records from it, and then learn the structure and the parameters of the BN from the simulated dataset.



Exercise3: Manual bn_template

Complete `bn_template.py` to calculate the probabilities for the Bayesian network below.



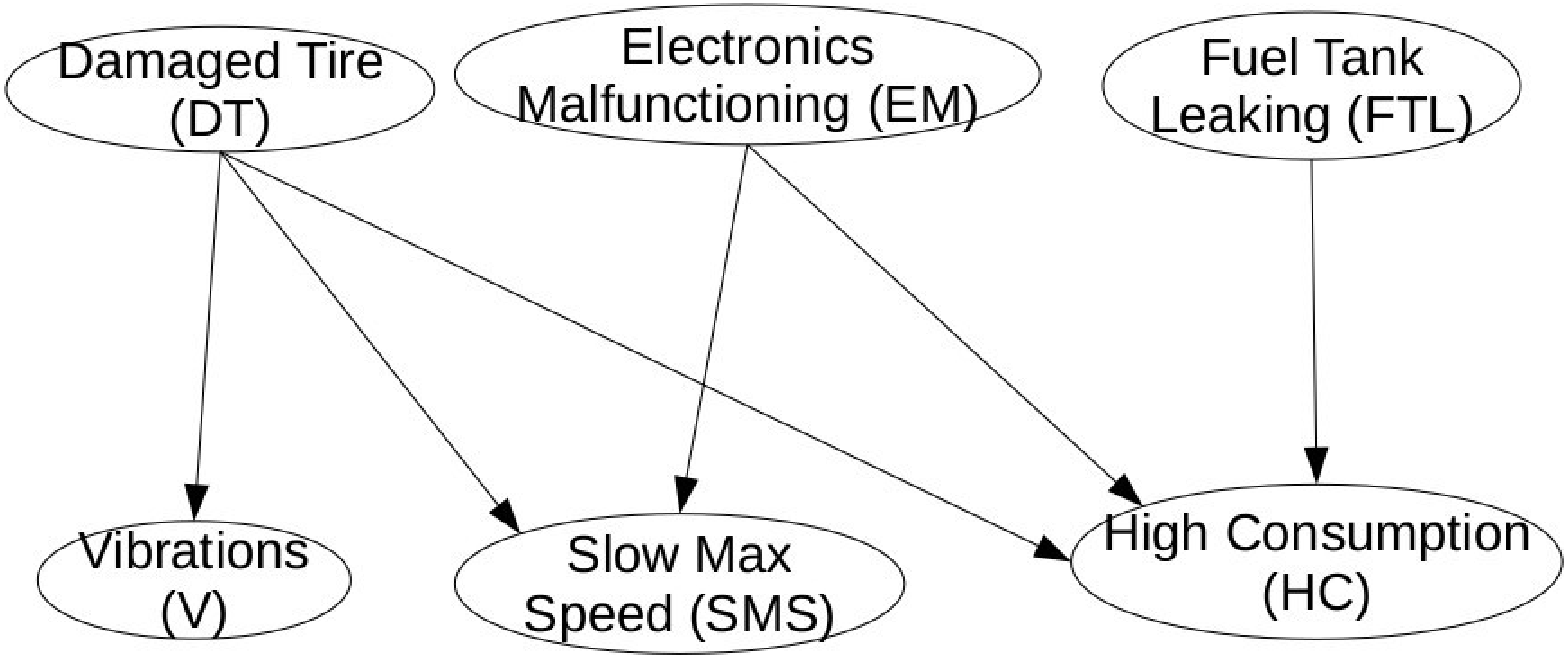
Homework

Task 1:
Modify the Bayesian network program from exercise 3 to calculate the probabilities for the network on the following slides.

Task 2:
You are interested in finding out the cause of a malfunction of your car.

You suspect a fault in your car, but you do not know in which component. You have noticed that there are vibrations while driving and it is difficult to reach the top speed. Your car does not consume more than usual.

Homework



Damaged Tire (DT)
Electronics Malfunctioning (EM)
Fuel Tank Leaking (FTL)
Vibrations (V)
Slow Max Speed (SMS)
High Consumption (HC)

Homework

DT	P(DT)
T	0.3
F	0.7

FTL	P(FTL)
T	0.2
F	0.8

EM	P(EM)
T	0.3
F	0.7

DT	EM	P(SMS=T | DT,EM)
T	T	0.05
T	F	0.6
F	T	0.3
F	F	0.7

DT	FTL	EM	P(HC=T | DT,FTL,EM)
T	T	T	0.9
T	T	F	0.8
T	F	T	0.3
T	F	F	0.2
F	T	T	0.6
F	T	F	0.5
F	F	T	0.1
F	F	F	0.01

DT	P(V | DT)
T	0.7
F	0.1